Cloud AI Capabilities Overview: Enterprise Architecture

Navigating AWS Bedrock, Azure AI Foundry, and Google Cloud for production GenAI.

1. Workload Scope

Business Contract

Grounded Support FAQ

Identity & ACL Rules

Ticket Approval Gate

`Zero Platform Assumptions`

2. Dual-Lane Topology

Architectural Isolation

Offline Ingestion Lane

Online Request Lane

Security Policy Checks

`Ingest / Query Decoupling`

3. 3-Cloud Mapping

Service Translation

AWS Bedrock Stack

Azure Foundry Stack

Google Cloud Stack

`Neutral Capability Contract`

4. Defensible Choice

Evaluation Board

5-Question Framework

Full Bill TCO Modeling

Rejected Alternative

`Executive Justification`

Amazon Bedrock

Curated models (Claude, Nova)

Knowledge Bases & Guardrails

AWS IAM & VPC Native

Azure AI Foundry

10,000+ model catalog & OpenAI

AI Search Semantic Rerank

Entra ID & M365 Ecosystem

Google Cloud

Gemini native 1M-2M context

Vertex AI Search & Grounding

BigQuery & Lakehouse Native

CORE RULE

Same contract.

Three stacks.

Zero vendor lock.

Three Student Deliverables

What every engineering student must be able to design and defend by the end of this module.

Deliverable 1: Building Blocks

Identify Managed AI SaaS

Models, RAG, Agents, Evals, Traces, Governance

Mastery Verification Criteria:

- Deconstruct vendor catalogs into 6 capabilities
- Distinguish managed RAG from custom search
- Spot managed agent limits vs custom loops
- Identify the full 7-item production cost shape

`Outcome: Capability Fluency`

Deliverable 2: Cross-Cloud Map

1 Workload, 3 Realizations

AWS Bedrock, Azure Foundry, Google Cloud

Mastery Verification Criteria:

- Keep the identical business contract intact
- Map offline data ingestion across 3 clouds
- Map online query and human approval gates
- Enforce enterprise IAM & document ACLs

`Outcome: Multi-Cloud Portability`

Deliverable 3: Defend Choice

Architectural Defense

5-Question Framework & Trade-off Audit

Mastery Verification Criteria:

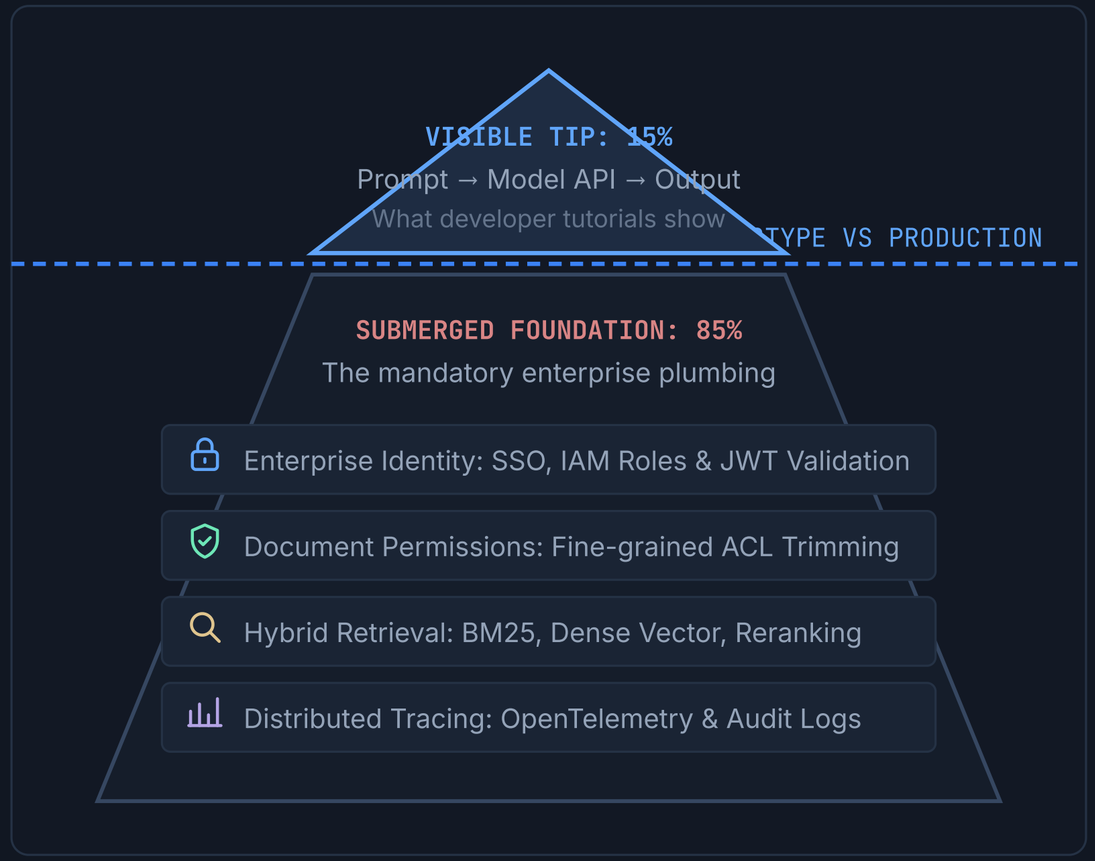
- Evaluate data boundary and compliance rules
- Calculate cost drivers beyond raw token price
- State exactly which platform was rejected
- Defend against team operational constraints

`Outcome: Executive Justification`

TAKEAWAY: A defensible decision requires trade-offs, explicit boundaries, and a rejected alternative.

Moving Beyond the Prototype: The Iceberg Problem

A raw model endpoint is only 15% of a production GenAI system.



Prototype Failures vs Production Reality

Failure Mode 1: Data Leaks & ACL Bypasses

Naive RAG indexes all corporate documents into a flat vector store.

Result: Standard users retrieve restricted executive payroll records.

Failure Mode 2: Unbounded Latency & Cost

Zero semantic caching, uncompressed system prompts, and loops spinning autonomously on ambiguous questions inflate bills 10x.

Failure Mode 3: Silent Quality Degradation

Document updates break table chunking: retrieval precision drops 30% with zero regression alerts or CI/CD test gates.

The Managed Cloud Advantage

Cloud platforms package IAM connectors, VPC isolation, vector indices, and automated evaluation so teams deploy securely in days, not quarters.

Focus engineering on business logic, approval gates & golden evals.

The 4-Phase System Design Flow

Reusing official cloud architecture training methodology: Concept to Decision.

PHASE 01 Concept

Mental Model

Define the workload contract
Establish security perimeter
Separate ingestion & inference
Human in the loop gates

KEY DELIVERABLE

Two-lane architecture
Grounded SLA definition
Strict permission policy

Neutral foundation

PHASE 02 Capability Map

The 6 Pillars

Group vendor services
Models & Model Garden
Knowledge & Hybrid Search
Agents, Evals & Observability

STANDARDIZATION

Platform capability matrix
Service classification
Bypass marketing hype

Universal language

PHASE 03 Guided Build

1 Workload × 3 Clouds

AWS Bedrock implementation
Azure Foundry implementation
Google Cloud implementation
Map concrete endpoints

SYSTEM VERIFICATION

Console topology translation
VPC & IAM boundaries
Action gate wiring

Equal business contract

PHASE 04 Decision

Defense Framework

5-question rubric
Full bill monthly TCO
Operational team profile
Rejected alternative defense

EXECUTIVE OUTPUT

Defensible recommendation
Documented trade-offs
Disqualified vendor proof

Boardroom ready

The 6 Pillars of Managed Enterprise AI

The standardized capability framework used to evaluate every cloud provider.

1. Foundation Models

Multi-provider catalog & open-weight weights
Serverless pay-per-token vs Provisioned Units
Adapter fine-tuning (LoRA) & distillation

`AWS Bedrock • Foundry • Model Garden`

2. Knowledge & RAG

Managed connectors (S3, SharePoint, Drive)
Automated chunking & embedding pipelines
Hybrid lexical + dense search & reranking

`Bedrock KB • AI Search • Vertex Search`

3. Agents & Orchestration

Stateful session memory & conversational state
Sandbox code interpretation & API invocation
Human in the loop approval confirmation gates

`AgentCore • Foundry Agent • Gemini ADK`

4. Evaluation & Benchmarks

Pre-release golden Q&A test sets
Automated Model-as-a-Judge scoring
Groundedness, relevance & toxicity metrics

`Bedrock Evals • Foundry Evals • Vertex Eval`

5. Tracing & Telemetry

Distributed spans via OpenTelemetry standard
Per-request token accounting & latency tracking
Failure clustering & error categorization

`CloudWatch/X-Ray • App Insights • Trace`

6. Enterprise Governance

IAM roles, SSO & document-level ACL filtering
Customer Managed Encryption Keys (CMEK)
Content safety, PII redaction & jailbreak filters

`Guardrails • Content Safety • Model Armor`

TAKEAWAY: Every vendor provides these 6 pillars: the difference lies in integration depth and operational friction.

The Benchmark Case: Grounded Policy Assistant

A single real-world workload that exposes every critical enterprise boundary.

USER QUERY (Sarah, Engineering):

"How long can I borrow a laptop, and where do I collect it?"

Context: Needs loaner equipment for client deployment

GROUNDED ASSISTANT RESPONSE:

"Standard loans are up to 14 days. Collect at IT Helpdesk B2."

Citation: Equipment Loan Policy (2025 rev), Section 2.1

Rule 1: Grounded Citation

No Evidence = No Answer

Must cite exact document name, section n

Pure parametric model knowledge is expli

FAIL CONDITION
Hallucinated collection hours

Rule 2: Identity & ACLs

Role-Based Trimming

Assistant can only search documents user

Enforces Entra ID / IAM groups prior to LLI

SECURITY CHECK
Pass JWT claims to vector filter

Rule 3: Action Approval

Human Confirmation

Can draft a ServiceNow ticket, but cannot

Requires explicit user UI confirmation befo

APPROVAL GATE
Prevents unintended actions

Rule 4: Audit & Trace

Zero Black-Box Runs

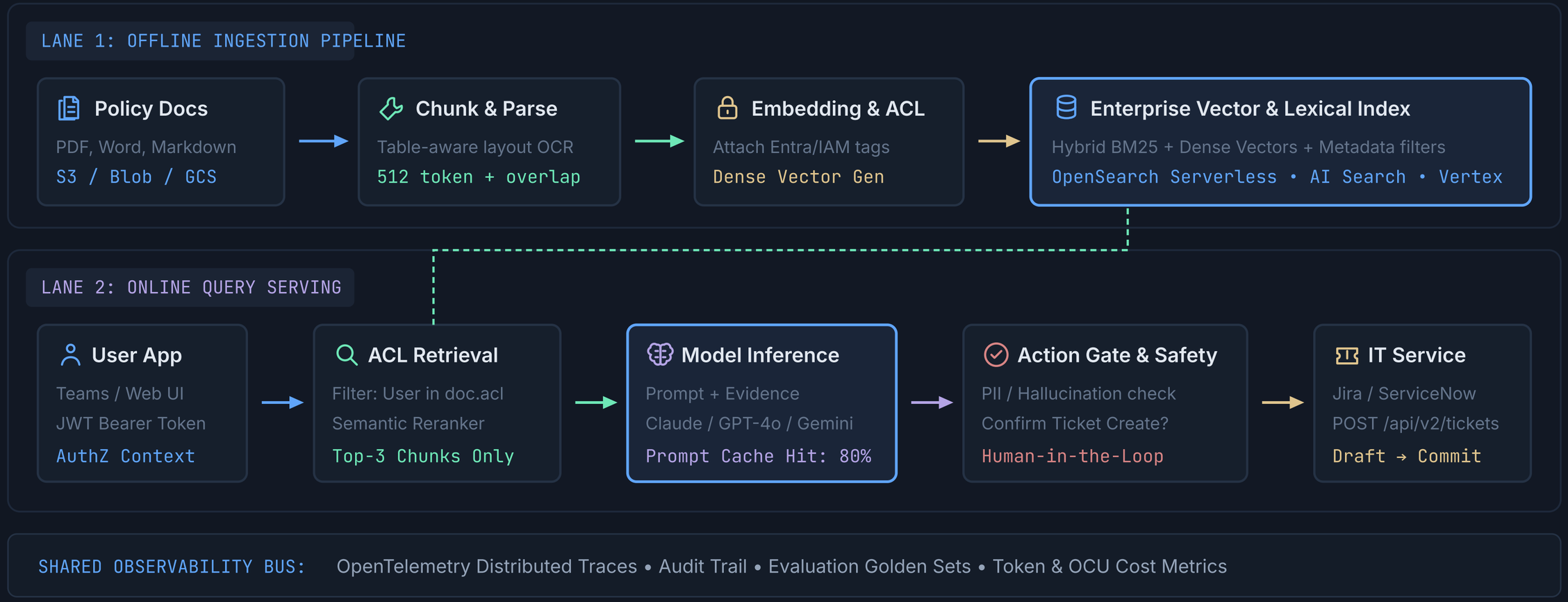
Full trace captured: prompt, retrieved chunks, t

Logged to centralized observability for compliа

OTEL AUDIT LOG
Immutable compliance store

Dual-Lane Architecture: Offline Ingestion vs Online Request

Decoupling continuous document indexing from real-time low-latency inference.



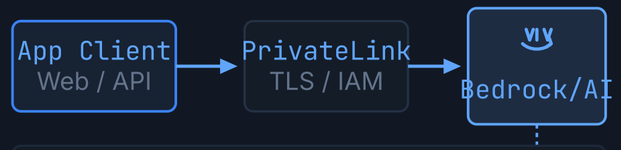
TAKEAWAY: Never mix document indexing pipelines with synchronous user inference paths.

The Tri-Model Deployment Spectrum

Selecting the operational model based on data classification, volume, and team maturity.

Managed Cloud AI

TOPOLOGY: MANAGED VPC ENDPOINT



Serverless Vector RAG + Guardrails
Zero GPU provisioning • Variable per-token

ENGINEERING METRICS

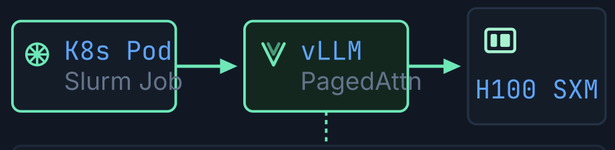
- Ops Overhead: 0 GPU ops or driver patching
- Time to Market: Days (instant serverless APIs)

OPTIMAL PRODUCTION FIT

Enterprise apps with variable traffic, strict enterprise SSO/IAM needs, and lean platform engineering teams.

Local / Self-Hosted

TOPOLOGY: AIR-GAPPED GPU CLUSTER



HF Local Weights + Private Vector DB
100% air-gap data sovereignty • 0 egress

ENGINEERING METRICS

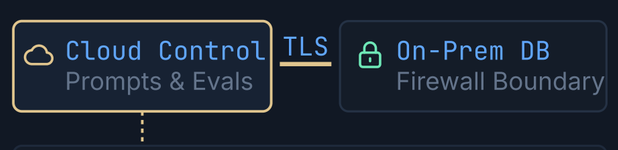
- Ops Overhead: 24/7 CUDA drivers & KV cache ops
- Economics: Fixed CapEx at >50M tok/day scale

OPTIMAL PRODUCTION FIT

Regulated defense or banking air-gaps, or steady ultra-high volume workloads with dedicated GPU infrastructure teams.

Hybrid Architecture

TOPOLOGY: SPLIT-PLANE ORCHESTRATION



Local Embeddings + Sanitized Cloud LLM
PII stripped before transit • Data stays local

ENGINEERING METRICS

- Security: Core records stay on-premise
- Network: Dedicated DirectConnect / ExpressRoute

OPTIMAL PRODUCTION FIT

Enterprises with heavy database gravity on-premise wanting rapid cloud GenAI rollout without migrating petabytes of core storage.

Model Catalog Philosophy: Curated vs Marketplace vs First-Party

Raw model counts are marketing signals: evaluate operational fit and regional availability.

Microsoft Azure AI Foundry

CATALOG PHILOSOPHY
Marketplace Hub • 10,000+ Models

OpenAI: GPT-4o • o1 • o3-mini
Flagship Strategic Exclusive Partner

Hugging Face: Direct Open Catalog

Mistral Large | Llama 3.3 70B

ENGINEERING EVALUATION

- API: Native OpenAI wire-format SDK
- SLAs: Tier-1 on OpenAI; OSS varies
- Privacy: Zero Data Retention via BAA
- Quota: PTU reservations for TPS SLA

OPTIMAL STACK FIT
OpenAI models & OSS marketplace

Amazon Bedrock

CATALOG PHILOSOPHY
Curated Multi-Provider Serverless

Anthropic: Claude 3.5 Sonnet / Haiku
Flagship Reasoning & Coding Engine

Amazon Nova Pro • Titan Embeddings V2

Llama 3.3 70B | Cohere Embed

ENGINEERING EVALUATION

- API: 100% Unified Converse API contract
- SLAs: Uniform AWS enterprise SLA
- Privacy: Zero Data Retention by default
- Quota: Serverless On-Demand + Provisioned

OPTIMAL STACK FIT
Claude 3.5 & unified Converse contract

Google Cloud Model Garden

CATALOG PHILOSOPHY
First-Party Multimodal & Open Models

Gemini: 2.0 Flash • 1.5 Pro
1M to 2M Token Native Multimodal Context

Claude 3.5 Sonnet + Gemma 2 (9B/27B)

Llama 3.3 70B | Mistral Large

ENGINEERING EVALUATION

- Context: Unmatched 1M-2M context tokens
- Multimodal: Audio, video, and PDF native
- TPU v5e: High-throughput Flash economics
- Caching: 75% savings via context caching

OPTIMAL STACK FIT
Native audio/video & 1M+ context

TAKEAWAY: Catalog size reflects ecosystem strategy: production stability requires SLAs, regional latency, and data privacy guarantees.

Qualitative Feature Matrix for the Support Assistant

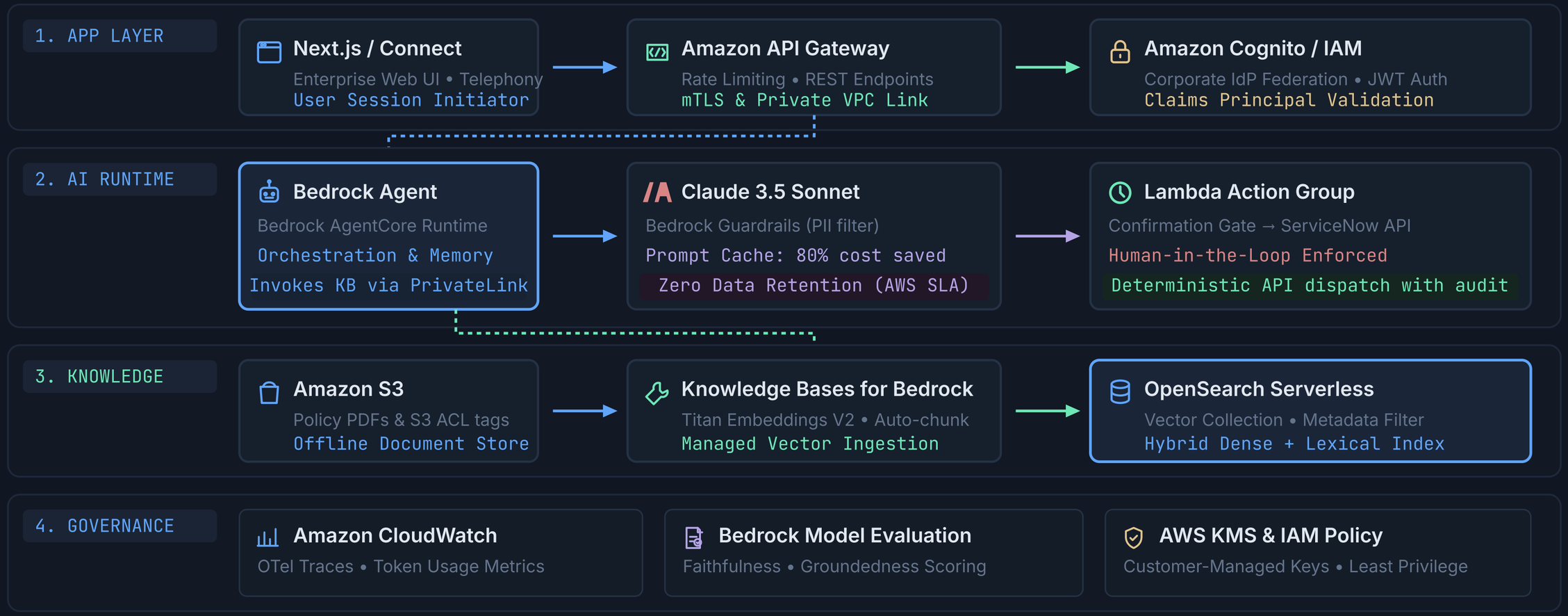
Evaluating AWS, Azure, and Google Cloud across 8 enterprise dimensions.

ARCHITECTURAL DIMENSION	AWS BEDROCK	AZURE AI FOUNDRY	GOOGLE CLOUD
1. Frontier Model Access	Strong (Claude, Nova)	Differentiator (OpenAI)	Strong (Gemini 2.0)
2. Managed RAG & Reranking	Baseline (Bedrock KB)	Differentiator (AI Search)	Strong (Vertex Ground)
3. Agent Runtime & Tools	Strong (AgentCore)	Baseline (Foundry Agent)	Strong (Gemini ADK)
4. Automated Evaluations	Baseline (Bedrock Eval)	Strong (Foundry Evals)	Strong (Vertex GenAI)
5. Enterprise Identity & ACLs	Strong (IAM Roles)	Differentiator (Entra ID)	Baseline (Cloud ID)
6. Data Platform Co-location	Strong (S3 • OpenSearch)	Strong (Fabric • Cosmos)	Differentiator (BigQuery)
7. Organizational Gravity	Native for AWS Estates	Native for M365 Shops	Native for Data / GCP
8. Long-Context & Caching	Prompt Cache (Claude)	Prompt Cache (OpenAI)	Differentiator (2M Cache)

TAKEAWAY: No single cloud wins every category: the optimal choice is anchored to where enterprise data and identity already live.

AWS Implementation: Bedrock-Centered Stack

How AWS services implement the two-lane support assistant topology.



TAKEAWAY: AWS provides a managed serverless AI runtime when coupled with existing AWS IAM and VPC infrastructure.

When AWS Bedrock is the Strongest Choice

Strategic alignment, architectural advantages, and operational gotchas.

Why Choose Amazon Bedrock?

1. Existing AWS Data Gravity

Enterprise policy documents, wikis, and databases already reside inside S3, RDS, or DynamoDB with mature IAM policies.

2. Enterprise Claude via Private VPC

Direct access to Anthropic Claude 3.5 Sonnet without public internet egress, routed through private AWS VPC endpoints.

3. Serverless Operational Model

Zero container clusters or server provisioning: Bedrock handles auto-scaling, cold-starts, and multi-AZ operational redundancy.

ORGANIZATIONAL ALIGNMENT

Strongest when the company is already an AWS shop: security, DevOps, IAM roles, and procurement pipelines are already in place.

Watch-Outs & Operational Gotchas

1. OpenSearch Serverless Cost Floor

Requires a minimum of 4 OCUs (2 search + 2 indexing), creating a fixed cost floor of ~$700/mo even for low-traffic PoCs.

2. Knowledge Bases Chunking Rigidity

Standard chunking splits complex PDF tables poorly; custom chunking requires building an external Lambda preprocessing step.

3. Cross-Region Routing & Quotas

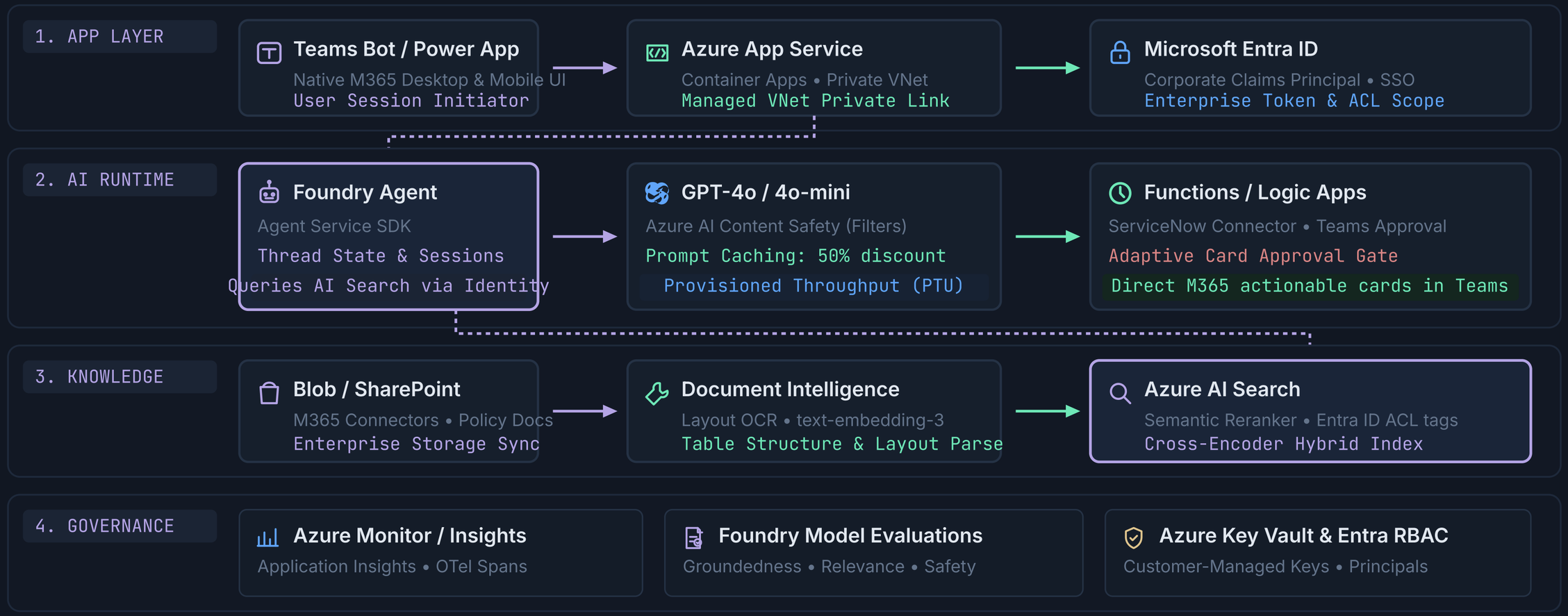
Claude 3.5 capacity in US-East-1 requires setting up cross-region inference profiles to avoid 429 Throttling exceptions.

ARCHITECTURAL MITIGATION

For pilot workloads under 10k queries/month, use Amazon Aurora PostgreSQL with pgvector instead of OpenSearch Serverless.

Azure Implementation: Foundry & AI Search Stack

Deploying the support assistant on the Microsoft enterprise ecosystem.



TAKEAWAY: Azure AI Foundry delivers the tightest integration with Microsoft 365, SharePoint, and corporate Entra ID.

When Azure AI Foundry is the Strongest Choice

Corporate identity fit, M365 synergy, and architectural trade-offs.

Why Choose Azure AI Foundry?

1. Native Microsoft Entra ID Trimming

Pass corporate JWT tokens directly into Azure AI Search: automatically enforces SharePoint document ACLs without custom sync scripts.

2. Enterprise OpenAI Models & SLAs

Priority access to GPT-4o, o1, and o3 with strict data guarantees: zero training on customer data and private virtual network endpoints.

3. Single Pane of Glass in Foundry

Unified portal consolidating model catalog, prompt playground, agent workflows, automated evaluations, and App Insights tracing.

ORGANIZATIONAL ALIGNMENT

Strongest for M365 enterprises: lowest friction path to launch grounded assistants inside Teams and Power Platform environments.

Watch-Outs & Operational Gotchas

1. Brand & Portal Renaming Churn

Fast product renaming (Azure OpenAI → AI Studio → AI Foundry) creates portal navigation churn and Python SDK version drift.

2. PTU Booking & Regional Quotas

Provisioned Throughput Units (PTUs) carry high monthly financial commitments: pay-per-token tier subject to strict TPM quota limits.

3. Azure AI Search Service Tier Costs

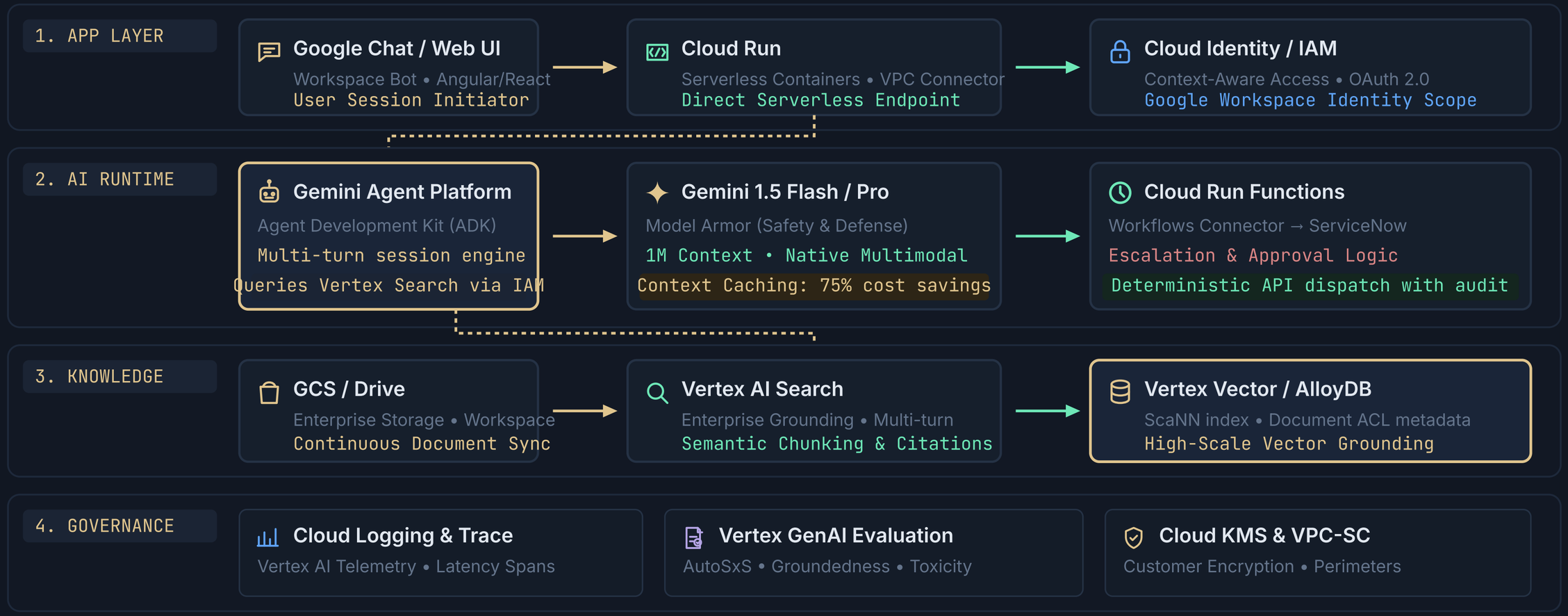
Semantic search features require Standard (S1) tier or higher, creating a ~$250/mo baseline cost before query volume ramps up.

ARCHITECTURAL MITIGATION

Pin exact SDK versions (azure-ai-projects) and utilize Global Standard deployments to maximize throughput elasticity.

Google Cloud Implementation: Gemini Enterprise Stack

Orchestrating the assistant with Gemini Enterprise Agent Platform and Vertex AI.



TAKEAWAY: Google Cloud excels in long-context document grounding, multimodal understanding, and data-warehouse co-location.

When Google Cloud is the Strongest Choice

Analytics synergy, context length supremacy, and platform evolution.

Why Choose Google Cloud?

1. BigQuery Lakehouse Co-location

If enterprise operational tables, customer service logs, and analytics live in BigQuery, Vertex AI provides zero-ETL grounding.

2. 1M-2M Native Context & Caching

Gemini native context windows allow entire equipment manuals or codebases to be cached in memory with up to 75% cost reduction.

3. Superior Flash Tier Unit Economics

Gemini 1.5 Flash delivers sub-second TTFT at pricing dramatically below comparable enterprise reasoning models.

ORGANIZATIONAL ALIGNMENT

Strongest for data-engineering and analytics-heavy firms that already run their primary data warehouse on Google Cloud Platform.

Watch-Outs & Operational Gotchas

1. Rapid Product Rebranding Pace

Transitions from Duet AI to Gemini Enterprise to Vertex Search create documentation churn and API parameter deprecations.

2. Regional Parity & Model Rollout

New Gemini checkpoints deploy first to us-central1; European and Asian regions often lag by several weeks or months.

3. Active Directory / Entra ID Friction

Federating legacy Windows Active Directory into Cloud Identity requires building and maintaining SAML/SCIM identity bridges.

ARCHITECTURAL MITIGATION

Deploy Workload Identity Federation for clean token exchange and explicitly lock model endpoint versions in Terraform.

Side-by-Side Enterprise Component Alignment

Translating architectural functions across AWS, Azure, and Google Cloud.

ARCHITECTURAL FUNCTION	AMAZON BEDROCK	AZURE AI FOUNDRY	GOOGLE CLOUD (GEMINI / VERTEX)
Model Catalog & API	Bedrock Multi-provider • Converse	Foundry 10k+ • Azure OpenAI	Model Garden • Gemini 1.5/2.0 API
Managed Knowledge & RAG	Knowledge Bases for Bedrock	Azure AI Search (Semantic Rerank)	Vertex AI Search & Enterprise Grounding
Agent & Tool Orchestration	Bedrock Agents • Bedrock AgentCore	Azure AI Foundry Agent Service	Gemini Agent Platform • ADK
Content Safety & Guardrails	Bedrock Guardrails (PII & Prompt)	Azure AI Content Safety	Model Armor • Vertex Safety Filters
Tracing & Observability	Amazon CloudWatch • AWS X-Ray	Azure Application Insights • OTel	Google Cloud Observability (Trace)
Model Evaluation	Bedrock Model Evaluation	Foundry Evaluation Hub	Vertex AI Model Evaluation
Enterprise Identity & Auth	AWS IAM • Amazon Cognito	Microsoft Entra ID (Native M365)	Cloud Identity • IAM Workload Auth

Enterprise Knowledge: Managed RAG vs Custom Hybrid Pipeline

Balancing development velocity against parsing quality and retrieval precision.

Path A: Turnkey Managed Cloud RAG (Black-Box Conveyor)

SPEED: 2 TO 5 DAYS TO DEPLOY

Cloud Storage
S3 • Blob • Drive
Auto-Sync Connector

→

Managed Ingestion Engine
Bedrock KB • AI Search • Vertex
Fixed-token (512/1000) chunking

→

Managed Vector Index
OpenSearch • ScaNN • AI Search
Zero Index Ops • Auto-scaled

→

RetrieveAndGenerate API
Single SDK call → Grounded response
Opaque relevance scoring

TRADEOFF: Standard fixed-token chunking cuts complex multi-row PDF tables and splits footnotes from headers.

Path B: Custom Modular Hybrid Pipeline (Production Precision)

VELOCITY: 4 TO 8 WEEKS BUILD

1. Layout & OCR
Docling • LlamaParse
Extracts tables to
clean Markdown tables

→

2. Table Chunker
Hierarchy • Headings
Keeps tables intact;
prepends breadcrumbs

→

3. Hybrid Store
Pinecone • Qdrant • Milvus
Dense Vector + Sparse BM25
Retrieves Top-50 candidates

→

4. Neural Rerank
Cohere Rerank v3.5
Cross-encoder reranking
Top-3 precision chunks

→

5. Synthesis
Target LLM
Accurate table
reasoning

PRODUCTION ARCHITECTURAL RULE: Start with Managed Cloud RAG to validate demand. Upgrade to Custom Pipeline when automated evaluations prove that retrieval failures (table extraction or cross-page correlation) violate business SLAs.

Agent Runtime Choices: When Do You Actually Need an Agent?

The spectrum of orchestration complexity: avoid over-engineering.

TIER 1 Deterministic DAG

TOPOLOGY: LINEAR PIPELINE

1. Query → Dense Embedding
2. Vector Search (Top-3)
3. Grounded Citation Output

Zero loops • 100% deterministic

PRODUCTION PROFILE

- Reliability: 99.5%
- Latency: Sub-1.2s TTFT
- Cost: 1x baseline
- Debug: Direct trace line

RECOMMENDED DEFAULT
80% of enterprise support

TIER 2 Prompt Agent

TOPOLOGY: FUNCTION ROUTER

1. LLM Tool-Call Selection
2. Host Executes Tool API
3. Formatted LLM Response

Single-turn branch • Fixed tools

PRODUCTION PROFILE

- Reliability: 95%
- Latency: 2.0s to 3.5s
- Cost: 2x baseline
- Debug: Standard API logs

BEST ARCHITECTURAL FIT
Single tool lookup (e.g. Weather)

TIER 3 Hosted Agent

TOPOLOGY: BOUNDED STATE MACHINE

Supervisor (Plan & Decide)
≤ 3 iterations
Worker Tool Execution
Session Memory & Gate

Bounded state loop • Max turns

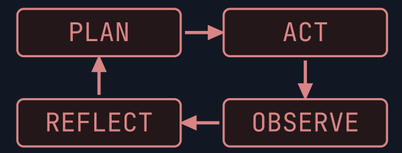
PRODUCTION PROFILE

- Reliability: 88%
- Latency: 4.0s to 8.0s
- Cost: 4x to 6x baseline
- Debug: Requires OTel spans

BEST ARCHITECTURAL FIT
Multi-system enterprise flows

TIER 4 Autonomous Loop

TOPOLOGY: UNBOUNDED CYCLE



Warning: Runaway loop risk

PRODUCTION PROFILE

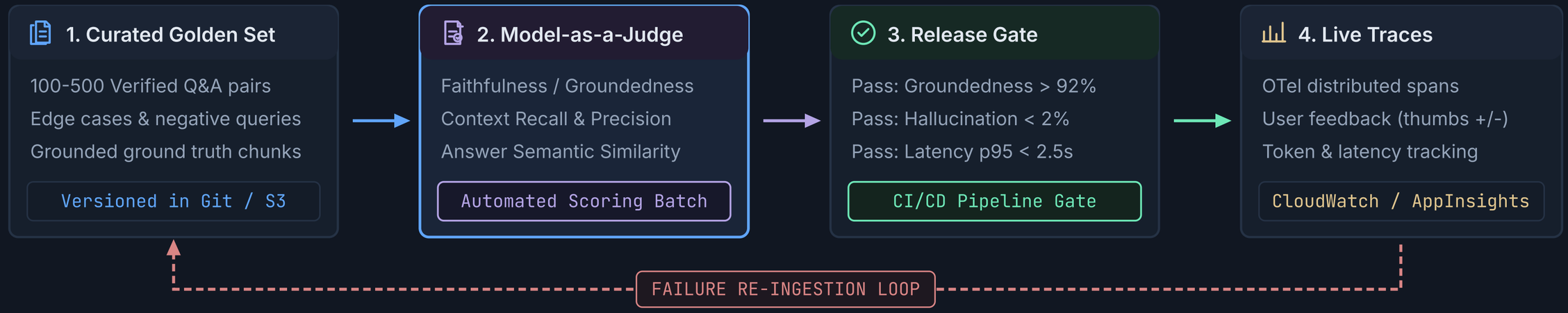
- Reliability: < 65% (Compounding)
- Latency: 10s to 60s+
- Cost: 10x to 20x+ baseline
- Debug: Non-deterministic

PRODUCTION WARNING
Avoid for end-user support

TAKEAWAY: For 80% of support assistants, a deterministic DAG is superior: reserve agentic loops for genuine multi-step workflows.

The Production Feedback Loop: Eval, Trace, & Mitigate

If you cannot trace a bad answer, you cannot operate an enterprise AI system.



THREE CRITICAL EVALUATION PILLARS (RAG TRIAD)

1. Groundedness (Faithfulness)

Is every generated claim directly supported by the retri

Catches model hallucinations and false inferences.

`Automated prompt-judge metric`

2. Context Relevance (Recall)

Did the retrieval step return the exact policy paragraph r

Catches poor chunking, outdated indexes, or bad embeddin

`Retrieval pipeline verification`

3. Answer Relevance

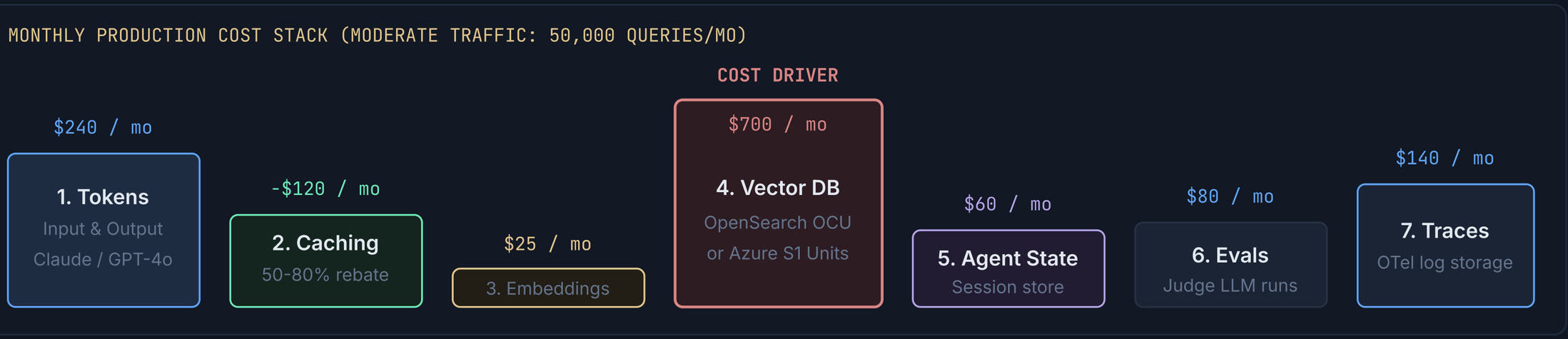
Does the synthesized text directly address the user inquiry?

Catches conversational drift and polite refusal boilerplate.

`End-to-end user satisfaction`

The Anatomy of an Enterprise GenAI Bill

Raw token price is only one component: dissecting the 7 production cost categories.



PRODUCTION COST REALITIES & LESSONS

1. Vector Minimums Dominate

OpenSearch Serverless (4 OCUs) or Azure AI Search (S1) cost $250-$700/mo.

Mitigate: Aurora pgvector for small apps

2. Prompt Caching Saves 50-80%

Static system instructions & policy chunks qualify for high-speed cache read rates.

Mitigate: Order prompt static-first

3. Tracing High-Frequency Overhead

Logging full prompt payloads to CloudWatch rapidly accumulates gigabytes of ingestion.

Mitigate: 10% sampling on successful runs

TAKEAWAY: Vector index minimums and high-frequency tracing often exceed raw model inference costs at moderate traffic.

Cost Optimization: Engineering Architectural Efficiency

Low cost is a design outcome, not a vendor discount: four engineering levers.

1. Context Compaction — 30-50% Savings

Trim Boilerplate Before Prompt Assembly

- Strip repetitive HTML headers, legal disclaimers, and conversational fluff.
- Clean markdown extraction preserves tabular structure in half the tokens.

```
Architecture: Client-side AST pruning pipeline
```

2. Semantic & Prompt Caching — 50-80% Savings

Bypass Model Inference on Repetitive Queries

- Exact match + cosine vector cache on Redis for common policy FAQs (0 tokens).
- Provider prompt caching (Bedrock Claude / Azure OpenAI / Gemini) on system prompts.

```
Architecture: Prefix caching with static system guidelines
```

3. Model Tiering & SLM Routing — 70% Savings

Route by Workload Difficulty

- Send 80% simple lookup questions to SLMs (Haiku / 4o-mini / Flash).
- Escalate multi-step exceptions to Frontier models (Sonnet / GPT-4o / Pro).

```
Architecture: Lightweight intent classification gateway
```

4. Batch & Flex Pricing — 50% Off-Peak Discount

Decouple Non-Interactive Inference

- Run offline document chunk summarization and embedding rebuilds overnight.
- Execute regression evaluation test batches via 24h batch APIs at 50% discount.

```
Architecture: Asynchronous SQS / Service Bus queues
```

The 5-Question Platform Selection Framework

The structured method for producing an executive-ready architectural recommendation.

Q1 — **Workload Pattern & Interaction Shape**
Is this a deterministic policy FAQ, complex multimodal document parser, or autonomous multi-step agent?
`Drives: Runtime Choice`

Q2 — **Data Gravity, Residency & Document ACLs**
Where do source documents currently reside (S3, SharePoint, BigQuery)? What are cross-border compliance boundaries?
`Drives: Cloud Gravity`

Q3 — **Model Capability Tier & Context Window**
Does the application require 1M+ context (Gemini), enterprise Claude (Bedrock), or OpenAI reasoning models (Foundry)?
`Drives: Model Selection`

Q4 — **Enterprise Identity & Observability Baseline**
Which identity provider (Entra ID vs AWS IAM) and tracing standard (OpenTelemetry) are approved by corporate SecOps?
`Drives: Governance Fit`

Q5 — **Operational Team Profile & Operator Burden**
Can the team maintain custom Kubernetes clusters and vLLM GPU servers, or does the workload require serverless SaaS?
`Drives: SaaS vs Self-Host`

`MANDATORY DEFENSE RULE: MUST EXPLICITLY NAME AND DISQUALIFY ONE REJECTED ALTERNATIVE`

The Multi-Cloud Demonstration Flow

Demonstrating one architectural contract across AWS, Azure, and Google Cloud consoles.

STAGE 01 Business Contract

- Review Sarah's laptop loan query
- Establish citation & ticket approval rule
- Define data boundary & user role

`Time Budget: 3 minutes`

STAGE 02 Neutral Topology

- Display two-lane reference diagram
- Point out offline vs online boundaries
- Keep diagram pinned on second monitor

`Time Budget: 4 minutes`

STAGE 03 AWS Bedrock

- S3 → Knowledge Base sync
- OpenSearch Serverless collection
- Bedrock Agent & Guardrails filter

`Console Demo: 6 minutes`

STAGE 04 Azure AI Foundry

- AI Search index & Entra ID ACL
- Foundry Agent Service wiring
- Semantic Reranker precision check

`Console Demo: 6 minutes`

STAGE 05 Google Cloud

- Vertex AI Search & Enterprise Grounding
- Gemini 1.5 Flash in Agent Platform
- Model Armor prompt injection test

`Console Demo: 6 minutes`

STAGE 06 Synthesis & TCO

- Open live pricing calculators
- Contrast OCU vs PTU vs PayG
- Announce defensible decision

`Decision Check: 5 minutes`

TAKEAWAY: Portals look different; the underlying identity, retrieval, and inference mechanisms are identical.

Design Task: Medical Claims Processing Assistant

Apply the 5-question framework to an end-to-end multimodal enterprise workload.

WORKSHOP SCENARIO PIPELINE

Input Claim
Scanned medical bill
PDF / Image JPEG

→

OCR Extract
Dates, Codes, Total
Multimodal / DocAI

→

Policy RAG
Match coverage
Vector Policy DB

Anomaly Reasoning
Flag billing outliers or duplicate codes
Frontier Model Synthesis

→

Human Review Gate
Confidence < 90% triggers nurse queue
Zero Blind Auto-Payment

ENTERPRISE CONSTRAINTS

- Strict HIPAA / GDPR: patient names & diagnoses must be redacted.
- Volume: 120,000 claims/month (peak between 8 AM and 4 PM).
- Core IT system: Existing insurance claims DB runs on Microsoft Azure.

Student Deliverable Scorecard

1. END-TO-END ARCHITECTURE DIAGRAM
Clear separation of ingestion OCR vs online validation.

2. PLATFORM SELECTION & RATIONALE
Concrete vendor choice defending data gravity & identity.

3. SECURITY & PII BOUNDARIES
De-identification filters and customer-managed KMS keys.

4. EVALUATION & REGRESSION TEST SETS
Golden test sets for code extraction accuracy & false claims.

5. MANDATORY DISQUALIFIED ALTERNATIVE
Explicit technical rationale why another cloud was disqualified.

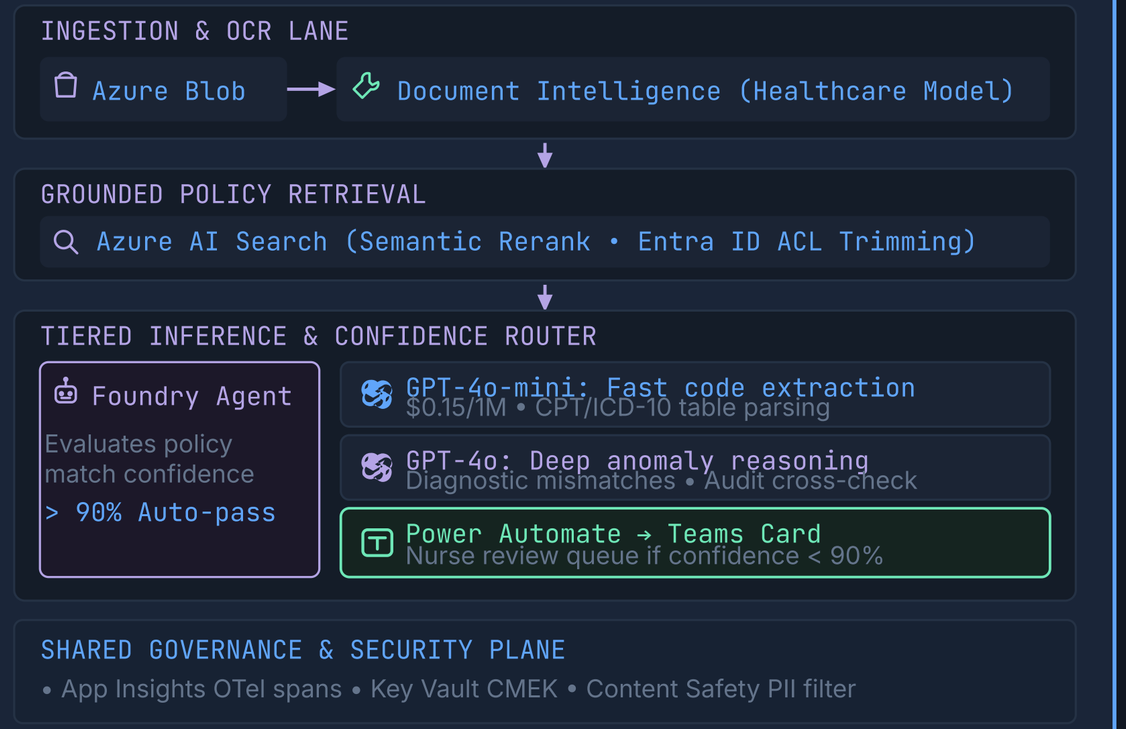
Session: 12 min build • 3 min defense

TAKEAWAY: 12 minutes to build, 3 minutes to defend: architecture under real enterprise constraints.

Reference Solution: Claims Assistant on Azure AI Foundry

What an executive-ready architecture decision package looks like.

Azure Reference Architecture (Healthcare BAA)



Executive Decision Defense

WHY AZURE WAS SELECTED (JUSTIFICATION)

- Organization already maintains BAA (HIPAA) on Azure.
- Claims reviewers work exclusively in Microsoft Teams & M365.
- AI Search semantic reranking delivers 94% recall on policy rules.
- Zero new procurement friction or vendor security audits required.

`Procurement Velocity: 0 Weeks delay (already active)`

EXPLICITLY REJECTED ALTERNATIVE: AWS BEDROCK

- Technical Capability: Bedrock with Claude 3.5 Sonnet is highly capable of medical extraction and complex reasoning.
- Rejection Reason 1: Introducing AWS requires new corporate SecOps security review and legal BAA procurement (adds 4-6 months delay).
- Rejection Reason 2: Lack of native Teams Adaptive Card approval integration creates unnecessary custom frontend maintenance.

`Disqualification: Organizational inertia > technical parity`

The Cloud AI Golden Rule

Managed platforms remove infrastructure work: they do not remove architecture decisions.

1. Evidence & Grounding

Where does the factual evidence live?

Never rely on model memory alone.

2. Identity & ACLs

Who is authorized to see each chunk?

Trim documents before prompt injection.

THE CLOUD AI GOLDEN RULE

Cloud platforms manage the models and hardware.
You own the boundaries, trust, and business risk.

3. Human Approval Gates

Which actions require human sign-off?

Zero blind ticket/payment execution.

4. Systematic Evaluation

How is quality continuously tested?

Golden sets and automated judges.

5. Observability & TCO

How are errors traced and bills tracked?

Vector minimums, OTel spans, tokens.